Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Agentic AI engineer specializing in **production LangGraph systems, RAG pipelines, and observability** - built AgentProbe, a multi-agent LangGraph system with end-to-end RAG, per-run trace enrichment, LLM-as-judge evaluation, and human-in-the-loop deployment gates. Also built *AI on FHIR*, an NLP-to-FHIR medical records query system with HIPAA-compliant RBAC - bringing healthcare data systems experience most agentic engineers don't have. 8+ years backend engineering at scale (1M+ records, 100K+ MAU, Python/AWS). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Multi-Agent Orchestration (LangGraph) | RAG Pipeline End-to-End | Agentic Observability & Tracing | Human-in-the-Loop Systems | LLM Evaluation (Task & Trajectory) | Healthcare Data Systems (FHIR) | LangChain / LangGraph | Context & Memory Management | Frontier Model Integration | Python / FastAPI | Large-Scale Data Pipelines | AWS / Docker / CI/CD

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **large-scale data processing frameworks** on AWS (S3, Glue, EC2) handling **1M+ records** with automated quality checks, Libsodium encryption, and zero security incidents - the production engineering discipline this role requires.
- Maintained **99.5% uptime** and sub-10 bug count for 6+ consecutive months via TDD/BDD quality gates and Docker containerization - cutting microservices execution time by **50%**.
- Built **20+ production modules and internal tools** (Python, TypeScript, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users** - strong practices in testing, CI/CD, versioning, and documentation.
- Improved production query performance by **30%** by architecting PostgreSQL schemas and data access layers across 20+ projects.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Delivered targeted Python and database mentoring for 30+ undergraduates through sprint-style feedback cycles.

PROJECTS

AgentProbe: Production Multi-Agent AI System OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · Ollama · React · Azure · Anthropic · OpenAI · Groq

Production agentic system built as my primary engineering project - not a prototype. Architecture: **LangGraph orchestration layer** with stateful branching, retries, and human-in-the-loop eval gates; **RAG Tracer agent** enriches every run with per-query trace data (retrieval scores, generation quality, anomaly flags, tool call logs); **5-metric LLM-as-judge engine** (correctness, hallucination, retrieval, completeness, compliance) evaluates at task and trajectory level; **Auto Fix agent** autonomously investigates eval failures, rewrites prompts, re-runs eval - human oversight gates fire at confidence thresholds; **React observability dashboard** with full trace visualization; **15-endpoint FastAPI backend**; multi-provider frontier model support (Anthropic, OpenAI, Groq, Ollama). Open-source, documented API, runs 100% locally. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP-Driven Medical Records Query System HEALTHCARE

Flask · LLMs · FHIR R4 · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

NLP pipeline mapping natural language queries (EN/ES) to structured FHIR R4 APIs via LLMs, BERT, and embeddings. HIPAA-compliant RBAC, containerized, structured output grounding reduces hallucination risk in clinical context. **Reduced retrieval time ~60%**. Rare combination of agentic AI engineering and healthcare data systems. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive Modeling on 100K+ Records

Python · R · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Full ML lifecycle on UK road accident data. **85%+ accuracy**; interactive dashboard for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

Agentic AI: LangChain, LangGraph, Multi-Agent Orchestration, RAG Pipelines, ChromaDB, Context Management, LLM Evaluation, LLM-as-Judge, Observability & Tracing, Human-in-the-Loop, Ollama, Anthropic, OpenAI, Groq, FastAPI, Prompt Engineering

Healthcare: FHIR R4, HIPAA-compliant RBAC, Clinical NLP, BERT, spaCy, Embeddings, Hugging Face

ML Stack: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Feature Engineering

Data & Cloud: AWS (S3, EC2, RDS, Glue), Azure, PostgreSQL, MySQL, NoSQL, Docker, CI/CD (GitHub Actions), RabbitMQ

Testing & Quality: TDD/BDD, Integration Testing, Selenium, Git, JIRA, Postman, Agile/Scrum